

MASTER 1 INTERNSHIP REPORT

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Bayesian inference and variant identification in meta-genomic data

Subsampling methods for massive datasets

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1 Introduction

Antibiotic resistance is a major challenge in modern medicine, underscoring the need to understand how resistance genes are transmitted and maintained within bacterial populations. In this project, we consider resistance genes found in livestock microbiota, where monitoring their diversity and abundance is essential for improving surveillance and controlling their spread.

High-throughput sequencing technologies produce millions of short DNA fragments, called reads (around 150 nucleotides). These reads are aligned to reference antibiotic resistance genes (1000–3000 nucleotides) to detect the presence of specific resistance genes. However, a single gene can exist in several genetic variants. A variant corresponds to a specific nucleotide sequence of a gene, differing from other variants by a set of mutations accumulated over time. In a metagenomic sample, several variants of the same gene may coexist. Since sequencing reads cover only short regions of the gene and variants are often very similar, they cannot be directly distinguished from the sequencing data.

One possible approach to reconstruct these variants is to exploit information from several biological samples that share the same set of variants but differ in their relative abundances. By comparing these samples, differences in variant proportions provide additional information that helps disentangle the contribution of each variant and reconstruct its underlying sequence. The objective is therefore to reconstruct the variants present in biological samples and estimate their relative abundances.

From a statistical perspective, this problem can be formulated as inference in a hierarchical mixture model. The observations are discrete nucleotide sequences, while the latent variables include the unknown variant sequences, the read-to-variant assignments, and the sample-specific mixture proportions. The resulting posterior distribution is high-dimensional and exhibits strong dependencies between latent variables and model parameters, making both likelihood evaluation and posterior exploration computationally challenging.

A Bayesian framework for this problem was proposed in the DESMAN algorithm Quince et al. (2017), which relies on a hierarchical model with a factorised approximate likelihood and a Gibbs sampling algorithm for posterior inference to jointly estimate variant sequences and their abundances. Building upon this work, Daniel, a former intern who previously worked on this project, first investigated the simplified setting in which the set of all possible variant sequences is assumed to be known. In this case, the inference procedure accurately estimates the relative abundances of the variants together with the sequencing error rate, showing that the underlying statistical model performs well when the latent sequences are fixed.

However, extending the method to the more realistic setting where the variant sequences are unknown reveals two major limitations. First, the approximate likelihood used in the original approach does not fully exploit the information contained in the sequencing data and may therefore lead to suboptimal inference. Second, the Gibbs sampling algorithm struggles to efficiently explore the posterior distribution because of the strong dependence between the latent variant sequences, the read assignments, and the mixture proportions. As a consequence, the Markov chain exhibits poor mixing and slow convergence, while the computational cost rapidly increases for large sequencing datasets.

The objective of this internship was to continue this line of research by revisiting these inference procedures and developing computationally more efficient alternatives. In particular, I investigated several likelihood approximation and subsampling strategies to accelerate posterior inference while preserving statistical accuracy. I also explored alternative MCMC sampling strategies to improve posterior exploration in the unknown-variant setting. These developments were incorporated into different inference frameworks and evaluated on simulated and real sequencing datasets.

This report is organised as follows. Chapter 2 introduces the biological context and the sequencing data, Chapter 3 presents the probabilistic model, Chapter 4 describes the MCMC inference methods, Chapter 5 introduces the proposed likelihood approximations and subsampling strategies, and Chapter 6 concludes the report. Additional material, including notation tables,

mathematical derivations, and supplementary algorithms, is provided in Appendix 6.

2 Biological and Computational Context

This chapter introduces the biological concepts necessary to understand the problem studied in this work and describes the sequencing data available for inference.

2.1 Biological background

2.1.1 DNA, genes and sequencing reads

DNA stores the complete genetic information of an organism and consists of a long sequence of nucleotides taking values in the following alphabet: $\mathcal{A} = \{A, C, G, T\}$. A gene is a functional region of DNA that contains the information required to perform a specific biological function. We focus on antibiotic resistance genes (1,000-3,000 nucleotides) and analyse them independently.

Current high-throughput sequencing technologies produce millions of short DNA fragments (100-200 nucleotides), called *reads*, each representing a partial observation of the original gene. Moreover, the genomic positions covered by a read are not necessarily consecutive after the alignment process, so reads may contain gaps. Different reads generally overlap over common genomic regions, allowing complementary information from multiple reads to be combined in order to reconstruct the underlying gene sequence. Since sequencing technologies are not error-free, some observed nucleotides may differ from the true underlying DNA sequence. Consequently, the available data consist of noisy and incomplete observations.

2.1.2 Genetic variants

A given resistance gene may exist in several closely related genetic variants: different versions of the same gene that usually differ at only a small number of nucleotide positions while preserving the same biological function.

Several variants may coexist within the same biological sample, each with its own relative abundance. Because sequencing reads are much shorter than the genes from which they originate, it is generally impossible to determine directly which variant generated a particular read. Reconstructing these hidden variants from the observed reads constitutes the main inference problem addressed in this work.

2.2 Sequencing data

The sequencing data analysed in this work were obtained independently for several antibiotic resistance genes. Each gene, therefore, gives rise to a separate dataset, and inference is performed independently for every gene.

2.2.1 Dataset structure

Each observation of the dataset corresponds to a single nucleotide identified in a sequencing read. Consequently, a read spanning several genomic positions yields multiple observations in the dataset. An example of the sequencing dataset used in this study is shown in Figure 1.

Figure 1: Example of the sequencing dataset used throughout this analysis.

	Sample	Read	Position	Base
	<chr>	<chr>	<dbl>	<chr>
1	CTU_ACL	"V1:1:HVJMDSDXY:1:1106:18159:5165 "	1	A
2	CTU_ACL	"V1:1:HVJMDSDXY:1:1106:18159:5165 "	2	T
3	CTU_ACL	"V1:1:HVJMDSDXY:1:1106:18159:5165 "	3	G
4	CTU_ACL	"V1:1:HVJMDSDXY:1:1106:18159:5165 "	4	A
5	CTU_ACL	"V1:1:HVJMDSDXY:1:1106:18159:5165 "	5	C
6	CTU_ACL	"V1:1:HVJMDSDXY:1:1106:18159:5165 "	6	A
7	CTU_ACL	"V1:1:HVJMDSDXY:1:1106:18159:5165 "	7	A
8	CTU_ACL	"V1:1:HVJMDSDXY:1:1106:18159:5165 "	8	A
9	CTU_ACL	"V1:1:HVJMDSDXY:1:1106:18159:5165 "	9	A
10	CTU_ACL	"V1:1:HVJMDSDXY:1:1106:18159:5165 "	10	A
# i 2,486,750 more rows				

The SAMPLE column identifies the biological sample from which the read originates. The READ column uniquely identifies each sequencing read. The POSITION column indicates the genomic position on the reference gene where the nucleotide is observed, while the BASE column records the observed nucleotide (A, C, G or T).

2.2.2 Dataset size

The sequencing datasets considered in this work typically contain between 30 and 50 biological samples, between 2 and 5 million observations (each observation corresponding to a nucleotide observed at a given genomic position), and approximately 8,000 to 20,000 distinct sequencing reads. The size of these datasets makes direct likelihood evaluation computationally demanding, motivating the development of the likelihood approximation and subsampling strategies proposed in this work.

2.2.3 Computational environment

All the developments presented in this work were carried out in a Linux-based computing environment. The complete implementation of the proposed methods, together with all numerical analyses and figures, was developed in the R programming language.

Since the inference algorithms involve computationally intensive operations and long execution times, access to a high-performance computing (HPC) cluster was available throughout the project. This made it possible to submit computationally demanding R scripts to remote computing nodes and retrieve the results several hours later, allowing experiments that would have been impractical on a standard workstation.

3 Probabilistic model

This chapter introduces the probabilistic framework used throughout this work. We first define the mathematical notation and the unknown parameters of the model before describing the hierarchical data-generation process relating the latent genetic variants to the observed sequencing reads.

3.1 General notation

The main sets and quantities used throughout this work are:

- $\mathcal{V}$: set of genomic positions of the analysed gene;
- G : number of genetic variants;
- $\mathcal{R}$: set of sequencing reads;
- S : number of biological samples.

A complete notation table is provided in Appendix A.

3.2 Parameters to infer and prior distributions

To quantify the uncertainty associated with the inference, we adopt a Bayesian framework. This allows all unknown quantities to be modelled through prior and posterior distributions. The objective of the model is to jointly infer the unknown genetic variants, their relative abundances across biological samples, the sequencing error rate, and the latent allocation of reads to variants.

Variant representation. Each variant $g \in \{1, \dots, G\}$ is represented by its nucleotide sequence over the genomic positions $v \in \mathcal{V}$. We encode this sequence using the indicator variables

$$\tau_{g,v,a} = \begin{cases} 1, & \text{if nucleotide } a \text{ is present at position } v \text{ in variant } g, \\ 0, & \text{otherwise.} \end{cases}$$

Since each genomic position contains exactly one nucleotide: $\sum_{a \in \mathcal{A}} \tau_{g,v,a} = 1, \forall g, v$. When the variants are unknown, we assign a categorical prior distribution to each nucleotide state, $(\tau_{g,v,a})_{a \in \mathcal{A}} \sim \text{Categorical}(\theta_{g,v,1}, \dots, \theta_{g,v,4})$, where $\theta_{g,v,a}$ denotes the prior probability of observing nucleotide a at position v for variant g . Equal values for $\theta_{g,v,a}$ correspond to a uniform prior on all possible variants, but it is possible to introduce prior information coming from known variants in existing databases.

Mixture proportions. The relative abundances of the variants within each biological sample are represented by $\pi_{g,s}$, where $\pi_{g,s}$ denotes the probability that a read in sample s has been generated by variant g . For each sample, these proportions satisfy $\sum_{g=1}^G \pi_{g,s} = 1$. The prior distribution over the mixture proportions is defined as $\pi_{\cdot,s} \sim \text{Dirichlet}(\alpha)$, where $\alpha \in \mathbb{R}_{>0}^G$ is the vector of concentration parameters. This defines a probability distribution over the simplex in $\mathbb{R}^G$. Typically, one or two variants dominate in a metagenomic sample. This biological prior knowledge can be incorporated by choosing a small value for α .

Sequencing error rate. Sequencing errors are modelled through a global error rate $\varepsilon \in [0, 1]$, which represents the overall probability of observing an incorrect nucleotide. This parameter is then used to define the sequencing error probabilities as $\varepsilon_{a,a} = 1 - \varepsilon$ and $\varepsilon_{a,b} = \frac{\varepsilon}{3}$ for $a \neq b$, where $\varepsilon_{a,b}$ denotes the probability of observing nucleotide b when the true nucleotide is a . The prior distribution assigned to ε is $\varepsilon \sim \text{Beta}(a, b)$, which defines a probability distribution over the interval $[0, 1]$. A typical error rate for current sequencing technology is 10^{-3} , and the prior on ε can encode this information.

Latent read allocation. For each read $r \in \mathcal{R}$, we introduce a latent variable $Z_r \in \{1, \dots, G\}$, which indicates the variant from which the read originates. Given the sample of origin s_r and the mixture proportions, we assume $Z_r \mid \pi_{\cdot,s_r} \sim \text{Categorical}(\pi_{\cdot,s_r})$. The collection of all latent allocation variables is denoted by $Z = \{Z_r\}_{r \in \mathcal{R}}$.

Together, the inferred parameters are therefore $\{\tau, \pi, \varepsilon, Z\}$, where τ defines the unknown genetic variants, π their relative abundances across samples, ε the sequencing error rate, and Z the latent origin of each read.

3.3 Observed sequencing data

The observed sequencing reads are assumed to be generated according to a hierarchical mixture model.

Each read $r \in \mathcal{R}$ is associated with

- a known biological sample $s_r \in \{1, \dots, S\}$,
- a known set of covered genomic positions $\mathcal{V}_r \subseteq \mathcal{V}$,
- an unknown latent genetic variant.

The biological sample from which each read originates is known from the dataset, whereas the genetic variant responsible for generating the read is unknown and must be inferred from the observed sequencing information.

For each read r and each covered position $v \in \mathcal{V}_r$, the observed nucleotide is encoded using the indicator variables

$$X_{r,v,a} = \begin{cases} 1 & \text{if nucleotide } a \text{ is observed at position } v \text{ in read } r, \\ 0 & \text{otherwise.} \end{cases}$$

Since exactly one nucleotide is observed at each covered position, $\sum_{a \in \mathcal{A}} X_{r,v,a} = 1$.

Equivalently, the observation can be represented by the one-hot encoded vector $X_{r,v,\cdot} \in \{0, 1\}^4$, where $X_{r,v,\cdot} = (X_{r,v,A}, X_{r,v,C}, X_{r,v,G}, X_{r,v,T})$. Therefore,

$$X_{r,v,\cdot} = \begin{cases} (1, 0, 0, 0), & \text{if nucleotide } A \text{ is observed,} \\ (0, 1, 0, 0), & \text{if nucleotide } C \text{ is observed,} \\ (0, 0, 1, 0), & \text{if nucleotide } G \text{ is observed,} \\ (0, 0, 0, 1), & \text{if nucleotide } T \text{ is observed.} \end{cases}$$

3.4 Likelihood

In this section, we derive the likelihood function underlying the proposed Bayesian model. As the likelihood links the observed sequencing data to the unknown parameters, it constitutes the core of the inference procedure. Moreover, its formulation will play a central role in the methodological developments and computational improvements introduced throughout this work.

3.4.1 Conditional observation model

Conditionally on the fact that read r originates from variant g , the probability of observing nucleotide a at position v is:

$$\mathbb{P}(X_{r,v,a} = 1 \mid \tau_g, \varepsilon) = \sum_{b \in \mathcal{A}} \tau_{g,v,b} \varepsilon_{b,a}.$$

This reflects:

- The true nucleotide b encoded by $\tau_{g,v,b}$.
- The sequencing error process $\varepsilon_{b,a}$.

3.4.2 Read-level likelihood under a fixed variant

For a given read r , we collect all the observed indicator variables into the matrix

$$X_r = \begin{pmatrix} X_{r,v_1,A} & X_{r,v_1,C} & X_{r,v_1,G} & X_{r,v_1,T} \\ X_{r,v_2,A} & X_{r,v_2,C} & X_{r,v_2,G} & X_{r,v_2,T} \\ \vdots & \vdots & \vdots & \vdots \\ X_{r,v_{|\mathcal{V}_r|},A} & X_{r,v_{|\mathcal{V}_r|},C} & X_{r,v_{|\mathcal{V}_r|},G} & X_{r,v_{|\mathcal{V}_r|},T} \end{pmatrix}.$$

Assuming independence across positions within a read, the likelihood of read r given a fixed variant g is a product of categorical likelihoods with parameter modulated by sequencing noise:

$$\mathbb{P}(X_r = (X_{r,v,a})_{v \in \mathcal{V}_r, a \in \mathcal{A}} \mid \tau_g, \varepsilon) = \prod_{v \in \mathcal{V}_r} \prod_{a \in \mathcal{A}} \left(\sum_{b \in \mathcal{A}} \tau_{g,v,b} \varepsilon_{b,a} \right)^{X_{r,v,a}}.$$

This expression aggregates, for each genomic position, the contribution of the observed nucleotide through the emission probabilities induced by the true sequence τ_g and the sequencing error matrix ε .

3.4.3 Mixture formulation and marginal likelihood

Since the generating variant of each read is unknown, we marginalise over all possible variants. Conditioned on the sample s_r , the marginal likelihood of observing read X_r (integrated over Z_r) is given by:

$$\mathbb{P}(X_r \mid \tau, \varepsilon, \pi, s_r) = \sum_{g=1}^G \pi_{g,s_r} \mathbb{P}(X_r \mid \tau_g, \varepsilon).$$

This corresponds to a weighted mixture over all possible latent variants, where the weights are given by the mixing proportions π_{g,s_r} .

3.4.4 Full dataset marginal likelihood

Assuming conditional independence between reads given the parameters, the marginal likelihood of the full dataset $X = \{X_r\}_{r \in \mathcal{R}}$ is obtained as the product over all reads:

$$\mathbb{P}(X \mid \tau, \varepsilon, \pi) = \prod_{r \in \mathcal{R}} \mathbb{P}(X_r \mid \tau, \varepsilon, \pi_{\cdot, s_r}).$$

This quantity defines the global marginal likelihood of the model, combining:

- The mixture structure over variants π .
- The genome sequencing process governed by the sequencing error rate ε .
- The variant-specific sequences encoded by $\tau_1, \dots, \tau_G$, which define the nucleotide composition of each latent variant across genomic positions.

The complete marginal likelihood of our model can therefore be written as follows:

$$\mathbb{P}(X \mid \tau, \varepsilon, \pi) = \prod_{r \in \mathcal{R}} \sum_{g=1}^G \pi_{g, s_r} \prod_{v \in \mathcal{V}_r} \prod_{a \in \mathcal{A}} \left(\sum_{b \in \mathcal{A}} \tau_{g, v, b} \varepsilon_{b, a} \right)^{X_{r, v, a}}$$

3.4.5 Augmented (complete-data) likelihood conditional on Z

In contrast to the marginal likelihood above, where Z is integrated out, when conditioning explicitly on the latent read assignments $Z = \{Z_r\}_{r \in \mathcal{R}}$, the likelihood of read r originates directly from variant Z_r :

$$\mathbb{P}(X_r \mid Z_r, \tau, \varepsilon) = \mathbb{P}(X_r \mid \tau_{Z_r}, \varepsilon).$$

Consequently, the augmented (or complete-data) likelihood conditional on Z takes the factorised form:

$$\mathbb{P}(X \mid Z, \tau, \varepsilon) = \prod_{r \in \mathcal{R}} \mathbb{P}(X_r \mid \tau_{Z_r}, \varepsilon) = \prod_{r \in \mathcal{R}} \prod_{v \in \mathcal{V}_r} \prod_{a \in \mathcal{A}} \left(\sum_{b \in \mathcal{A}} \tau_{Z_r, v, b} \varepsilon_{b, a} \right)^{X_{r, v, a}}.$$

The joint probability distribution of the complete data (X, Z) is then given by:

$$\mathbb{P}(X, Z \mid \tau, \varepsilon, \pi) = \mathbb{P}(Z \mid \pi) \mathbb{P}(X \mid Z, \tau, \varepsilon) = \prod_{r \in \mathcal{R}} \pi_{Z_r, s_r} \mathbb{P}(X_r \mid \tau_{Z_r}, \varepsilon).$$

Distinguishing between these two formulations is essential:

- The **marginal likelihood** $\mathbb{P}(X \mid \tau, \varepsilon, \pi)$ is used when evaluating model likelihoods or log-likelihood values directly (e.g. for parameter evaluation or convergence monitoring).
- The **augmented likelihood** $\mathbb{P}(X \mid Z, \tau, \varepsilon)$ (and complete distribution $\mathbb{P}(X, Z \mid \tau, \varepsilon, \pi)$) is used within a Gibbs sampling algorithm (see details in Section 4) where Z is explicitly sampled as a parameter.

Remark. The inference problem considered in this work is finite-dimensional. However, the cardinality of the sets involved in the likelihood evaluation is extremely large in practice. Consequently, each likelihood evaluation entails a substantial computational cost. These computational difficulties will become particularly apparent when introducing the Markov Chain Monte Carlo (MCMC) algorithm in Section 4, and will be addressed through several likelihood approximation strategies presented in Section 5.

3.4.6 Log-likelihood and numerical underflow

The direct evaluation of the likelihood suffers from numerical underflow because it involves products of a large number of probabilities. In practice, all computations are therefore performed in the log-domain using the log-sum-exp trick. A detailed derivation of this reformulation is provided in Appendix B. The resulting expression used throughout this work is

$$\mathcal{L}(X \mid \pi, \tau, \varepsilon) = \sum_{r \in \mathcal{R}} \log \left(\sum_{g=1}^G \exp \left(\log \pi_{g, s_r} + \sum_{v \in \mathcal{V}_r} \sum_{a \in \mathcal{A}} X_{r, v, a} \log \left(\sum_{b \in \mathcal{A}} \tau_{g, v, b} \varepsilon_{b, a} \right) \right) \right)$$

3.4.7 Comparison with the DESMAN likelihood

Although the probabilistic framework presented in this work is inspired by the DESMAN model, the likelihood formulation implemented here differs from the original. Among other differences, DESMAN involves an approximation of our likelihood based on splitting reads into pseudo-reads of length one and collecting identical pseudo reads. This considerably reduces the dimensionality of the inference problem and the computational complexity of evaluating the likelihood, but it loses information about dependencies between some genome positions, which can be essential for reconstructing variant sequences. These differences are discussed in detail in Appendix C.

4 Posterior sampling for the exact model

We derive an MCMC scheme targeting the posterior distribution associated with the exact read-level likelihood. Its computational complexity is too large to use it on a large dataset, but it will be useful to assess the correctness of the approximate methods presented later.

4.1 Introduction to Markov Chain Monte Carlo

Markov Chain Monte Carlo (MCMC) methods are a class of stochastic algorithms for generating samples from complex probability distributions, particularly when direct sampling is not feasible.

Let $\pi(\theta)$ denote a target probability distribution, typically a Bayesian posterior distribution. The main idea of MCMC is to construct a Markov chain

$$\theta^{(1)}, \theta^{(2)}, \dots, \theta^{(T)}$$

whose stationary distribution coincides with $\pi(\theta)$ Hoff (2009).

A stochastic process $\{\theta^{(t)}\}_{t \geq 1}$ is a Markov chain if

$$p(\theta^{(t)} \mid \theta^{(t-1)}, \dots, \theta^{(1)}) = p(\theta^{(t)} \mid \theta^{(t-1)}).$$

MCMC methods construct a transition kernel $K(\theta' \mid \theta)$ such that the target distribution $\pi(\theta)$ is invariant:

$$\int \pi(\theta) K(\theta' \mid \theta) d\theta = \pi(\theta').$$

Under appropriate conditions, in particular if the chain is ergodic, the distribution of $\theta^{(t)}$ converges to the stationary distribution $\pi(\theta)$ as $t \rightarrow \infty$, independently of the initialisation of the chain. Consequently, long-run samples from the chain can be used to approximate expectations under $\pi(\theta)$:

$$\mathbb{E}_{\pi}[f(\theta)] \approx \frac{1}{T} \sum_{t=1}^T f(\theta^{(t)}).$$

4.2 Case 1: Known variants (fixed τ)

In this first case, we assume that the set of possible genetic variants is known beforehand. More precisely, the tensor τ , whose entries $\tau_{g,v}$ are vectors of length four indicating the nucleotide composition of variant g at genomic position v , is considered to be fixed and fully observed.

This assumption corresponds to the situation where a dictionary of candidate variants is available, for example from existing genomic databases or previous sequencing experiments. The observed data are therefore assumed to originate from one or several variants contained in this dictionary. If the dictionary contains variants that are not actually present in the sample, their estimated relative abundances are expected to be close to zero. This setting provides a simpler inference problem than when the variant sequences themselves must also be inferred. But the simplified approach can fail spectacularly if the dictionary is too large or incomplete. Building an adequate dictionary is itself a challenging bioinformatics problem.

4.2.1 Posterior distribution

Given the observed reads and the prior distributions introduced in the previous section, the objective is to sample from the joint posterior distribution

$$p(\boldsymbol{\pi}, \varepsilon, Z \mid X, \tau).$$

This posterior distribution combines the information provided by the observed sequencing reads with the prior knowledge encoded in the model. However, it does not admit a closed-form expression from which direct sampling is feasible.

4.2.2 Gibbs sampling

To approximate the posterior distribution, we use a Gibbs sampler. Gibbs sampling is a Markov Chain Monte Carlo (MCMC) method that constructs a Markov chain whose stationary distribution is the target posterior distribution. It is particularly well suited to our model because the full conditional distributions of all unknown variables can be derived analytically.

Starting from an initial configuration, the algorithm iteratively updates each group of variables by sampling from its full conditional distribution while keeping the remaining variables fixed. We therefore need to derive the following conditional posterior distributions:

$$p(Z \mid \boldsymbol{\pi}, \varepsilon, X, \tau), \quad p(\boldsymbol{\pi} \mid Z, \varepsilon, X, \tau), \quad p(\varepsilon \mid Z, \boldsymbol{\pi}, X, \tau).$$

4.2.3 Full conditional distributions

We now derive the conditional posterior distributions required by the Gibbs sampler.

Step 1: Latent assignments Z We sample each latent assignment Z_r from its full conditional posterior distribution

$$p(Z_r \mid \boldsymbol{\pi}, \varepsilon, X, \tau).$$

This distribution depends on the current estimates of the mixture proportions $\boldsymbol{\pi}$ and the observation model. It defines the probability that each read is assigned to each variant given all other parameters.

For each read r and variant $g \in \{1, \dots, G\}$, the full conditional posterior distribution simplifies to

$$p(Z_r = g \mid X, \tau, \boldsymbol{\pi}, \varepsilon) = \frac{\pi_{g,s_r} p(X_r \mid \varepsilon, \tau_g)}{\sum_{g'=1}^G \pi_{g',s_r} p(X_r \mid \varepsilon, \tau_{g'})}$$

The derivation of this full conditional distribution is provided in Appendix D.1.

Step 2: Mixture proportions $\boldsymbol{\pi}$ For each variant $g \in \{1, \dots, G\}$ and sample s , we define

$$n_{g,s} = \sum_{r \in \mathcal{R}: s_r = s} \mathbb{1}_{\{Z_r = g\}}.$$

The mixture proportions are sampled from $p(\boldsymbol{\pi}_{\cdot,s} \mid Z, \varepsilon, X, \tau)$.

Using Dirichlet-multinomial conjugacy, the corresponding full conditional distribution is

$$\boldsymbol{\pi}_{\cdot,s} \mid Z, \varepsilon, X, \tau \sim \text{Dirichlet}(\alpha + n_{1,s}, \dots, \alpha + n_{G,s})$$

The derivation of this full conditional distribution is provided in Appendix D.2

This distribution updates the relative abundances of the variants within each sample according to the current latent allocations.

Step 3: Sequencing error parameter ε

Define

$$M = \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}_r} \mathbf{1}\{x_{r,v} = t_{Z_r,v}\}, \quad \text{and} \quad E = \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}_r} \mathbf{1}\{x_{r,v} \neq t_{Z_r,v}\}.$$

Using Beta conjugacy, the conditional posterior distribution is

$$\boxed{\varepsilon \mid X, Z, \pi, \tau \sim \text{Beta}(a + E, b + M)}$$

The derivation of this full conditional distribution is provided in Appendix D.3

4.2.4 Implementation Overview

The Gibbs sampler is obtained by iteratively sampling from the three conditional posterior distributions derived above. At each iteration, the algorithm successively updates

$$Z \longrightarrow \pi \longrightarrow \varepsilon,$$

using the current values of the remaining variables. Under the conditions presented in the previous section, this procedure constructs a Markov chain whose stationary distribution is the target posterior distribution.

After a burn-in period, the generated samples can be treated as approximate draws from the posterior distribution and used for Bayesian inference.

Pseudo-code The Gibbs sampling algorithm, which I implemented in the R programming language, is summarised by the pseudocode provided in Appendix E.1.

4.2.5 Results

To evaluate the proposed inference framework, I consider a simulated dataset composed of 10 biological samples and 10 fixed genetic variants. Each variant consists of 801 nucleotides, and for each biological sample I generate approximately 300 sequencing reads, each covering 151 genomic positions. Since the true variant sequences, the mixture proportions and the sequencing error rate ε are known, this dataset provides a controlled setting in which I can assess the behaviour of the inference procedure.

More precisely, I analyse the following aspects:

- **Convergence towards high-likelihood regions.** Figure 2 shows the evolution of the log-likelihood throughout the MCMC iterations. I observe that all chains rapidly converge towards regions of high likelihood and, after a short burn-in period, remain close to the likelihood obtained using the true parameters employed to generate the simulated dataset.
- **Accuracy of the posterior estimates.** Figure 3 presents the posterior distributions of several representative parameters. I observe that the true parameter values belong to the corresponding posterior distributions and are accurately recovered by the posterior means. Moreover, the credible intervals contain the true values, indicating that the inference procedure successfully estimates the parameters used to generate the data.
- **Convergence of the Markov chains.** Figure 4 illustrates the evolution of the estimated mixture proportions over the MCMC iterations. The chains rapidly stabilise around the true abundances and exhibit only small fluctuations after convergence, providing evidence of satisfactory convergence of the Markov chains.

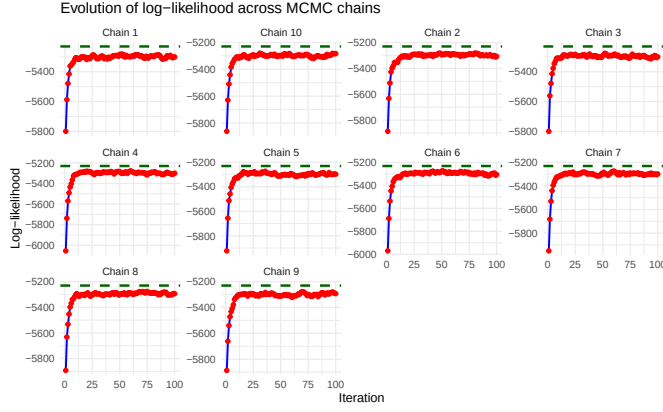


Figure 2: The x-axis represents the MCMC iterations, while the y-axis corresponds to the log-likelihood value computed using the parameters sampled at each iteration. The green line indicates the maximum expected log-likelihood, corresponding to the value obtained using the true parameters employed to generate the simulated dataset.

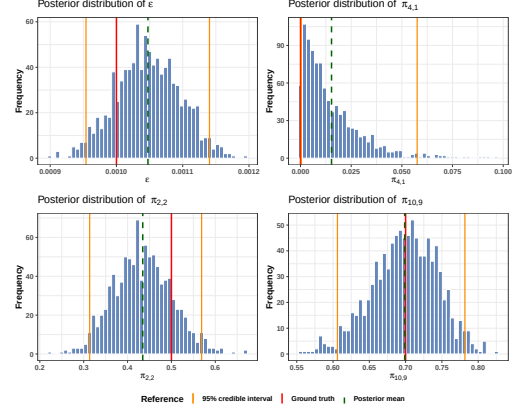


Figure 3: Posterior distributions of selected model parameters: sequencing error rate ε and mixture proportion parameters $\pi_{2,2}$, $\pi_{4,1}$ and $\pi_{10,9}$. Yellow lines indicate the 95% credible intervals, green dashed lines the posterior means, and red lines the true parameter values.

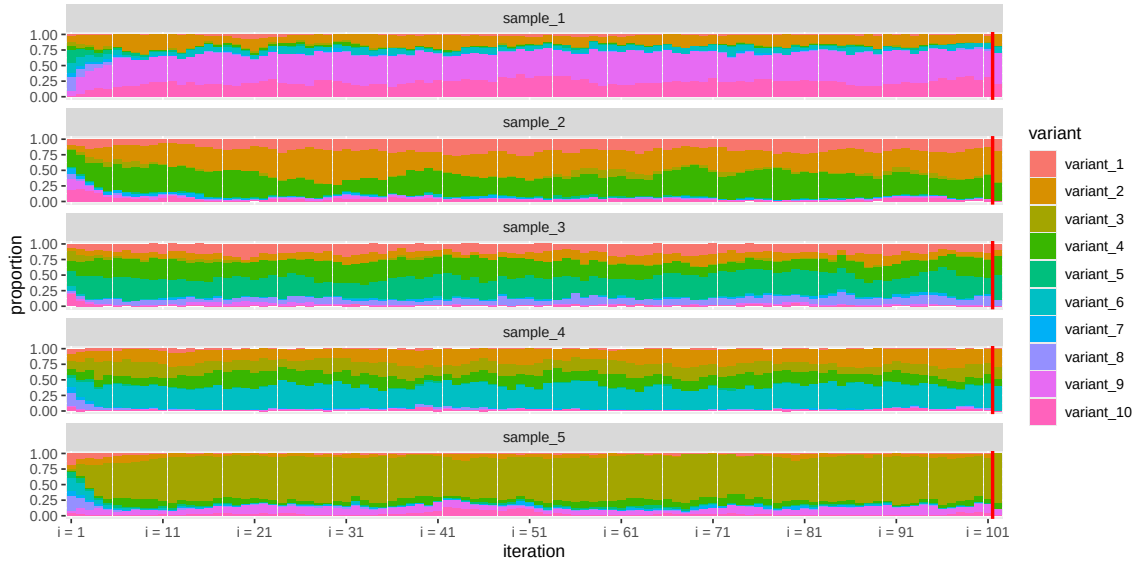


Figure 4: Evolution of the estimated mixture proportions (y-axis) across MCMC iterations (x-axis). Each panel corresponds to one biological sample, and each colour represents the estimated proportion of a given genetic variant. The final column, separated by the red vertical line, corresponds to the ground truth proportions used to generate the simulated dataset.

4.3 Case 2: Unknown Variant Sequences

We now consider the case where the variant sequences $\tau = (\tau_1, \dots, \tau_G)$ are no longer assumed to be known. Instead, they are treated as latent variables that must be inferred from the observed sequencing data.

Compared to the previous setting, this introduces an additional layer of uncertainty: the model must now explain the data without access to fixed reference sequences. In other words, the observed reads are no longer compared to known templates, but must be matched to sequences that are themselves unknown and must be reconstructed.

The key difficulty in this setting comes from the interaction between the latent assignments Z and the variant sequences τ . Although they are updated separately in the Gibbs sampling

procedure, they are strongly coupled through the likelihood. Indeed, the assignment of reads to variants depends on the current estimate of τ , while the reconstruction of each τ_g depends on the subset of reads currently associated to variant g through Z . This creates an iterative dependence between both components, which increases the complexity of the inference problem compared to the known-sequence case and may lead to poor mixing of the Markov chain, as updates of Z and τ can become highly correlated.

Remark. In practice, the number of variants G is also unknown. A common strategy in Bayesian mixture models is to deliberately overestimate the number of mixture components and let the posterior identify redundant ones. This approach is theoretically justified in the simpler setting of overfitted Gaussian mixture models, where extra components are asymptotically emptied or merged under suitable prior specifications Rousseau and Mengersen (2011). Following this principle, we set G as an upper bound on the number of variants and, after posterior sampling, merge identical components (same sequence τ_g) and sum their weights. An alternative approach would be to develop a repulsive mixture prior to force separation of the components, but this is out of the scope of this work.

Posterior Conditional distribution of τ . Given X , Z , and ε , the conditional distribution of each $\tau_{g,v}$ depends only on the reads currently assigned to variant g that cover position v .

For each variant g , position v , and nucleotide $b \in \mathcal{A}$, define

$$n_{g,v} = \sum_{r \in \mathcal{R}} \mathbf{1}\{Z_r = g, v \in \mathcal{V}_r\},$$

where $\mathcal{V}_r$ denotes the set of genomic positions covered by read r , and

$$m_{g,v,b} = \sum_{r \in \mathcal{R}} \mathbf{1}\{Z_r = g, v \in \mathcal{V}_r, X_{r,v} = b\},$$

which counts the number of reads assigned to variant g that observe nucleotide b at position v .

Then, for each nucleotide $b \in \mathcal{A}$,

$$\mathbb{P}(\tau_{g,v} = b \mid X, Z, \varepsilon) \propto \theta_{g,v,b} (1 - \varepsilon)^{m_{g,v,b}} \left(\frac{\varepsilon}{3}\right)^{n_{g,v} - m_{g,v,b}}.$$

Normalizing over $b \in \mathcal{A}$ yields

$$\tau_{g,v} \mid X, Z, \varepsilon \sim \text{Categorical}(p_{g,v,1:4}),$$

where

$$p_{g,v,b} = \frac{w_b}{\sum_{b' \in \mathcal{A}} w_{b'}}, \quad w_b = \theta_{g,v,b} (1 - \varepsilon)^{m_{g,v,b}} \left(\frac{\varepsilon}{3}\right)^{n_{g,v} - m_{g,v,b}}.$$

In practice, the variables $\tau_{g,v}$ may be sampled independently for all $g \in \{1, \dots, G\}$ and $v \in \mathcal{V}$, although this creates a poorly mixing Markov chain.

4.3.1 Gibbs Sampling with Unknown Variants

When the variant sequences τ are unknown, they are included as an additional step in the Gibbs sampler. The remaining updates for Z , π , and ε are unchanged from the known-sequence case, since they are always conditioned on the current value of τ .

The full Gibbs sampler iterates as follows:

1. Update $Z \mid \pi, \tau, \varepsilon, X$
2. Update $\pi \mid Z$
3. Update $\varepsilon \mid Z, \tau, X$
4. Update $\tau \mid X, Z, \varepsilon$

Pseudocode. Since the updates of Z , π and ε are identical to those of the Gibbs sampler for the known-variants case, we only detail the update of τ . The corresponding algorithm, which I implemented in the R programming language, is summarised by the pseudocode provided in Appendix E.2.

4.3.2 Results

To evaluate the inference procedure in the unknown-variant setting, I again consider the previously introduced simulated dataset. In the following figures, I compare inferred and true variants by matching the sequences that share the same informative positions (300, 350 and 400). This matching procedure makes it possible to track the estimated mixture proportions across the MCMC iterations and to compare them directly with the ground truth.

The main observations are summarised below:

- **Limitations of the Gibbs sampler.** In Figure 5, I compare the evolution of the estimated mixture proportions with the true proportions after matching the inferred variants to the true variants using their informative positions. In this case, I observe that the Gibbs sampler struggles to recover the true variants. During several stages of the sampling procedure, the chain assigns a substantial proportion to inferred variants that cannot be matched to any true variant, indicating that it becomes trapped in local modes of the posterior distribution. This behaviour is mainly explained by the strong dependence between the latent variables Z and τ , which severely limits the sampler’s ability to efficiently explore the posterior distribution.
- **Future directions.** These observations show that a straightforward extension of the Gibbs sampler developed for the known-variant setting is not sufficient in the unknown-variant case and motivate the development of more advanced sampling strategies. During this internship, I therefore investigated several alternative approaches, including *Block Sampling*, where several positions of the latent variant sequences are updated simultaneously, since changing only one position at a time is often insufficient to escape from local modes. I also investigated *Block-wise Sampling* and *Metropolis–Hastings* updates, all of which are designed to improve posterior exploration and help the Markov chain escape from local modes. Although there was not enough time to fully implement and experimentally evaluate these methods within the scope of this internship, I obtained promising results with split–merge Metropolis–Hastings moves, which represent promising directions for future work. Their main ideas and implementation details are presented in Appendix H.1, Appendix H.2, and Appendix H.3.

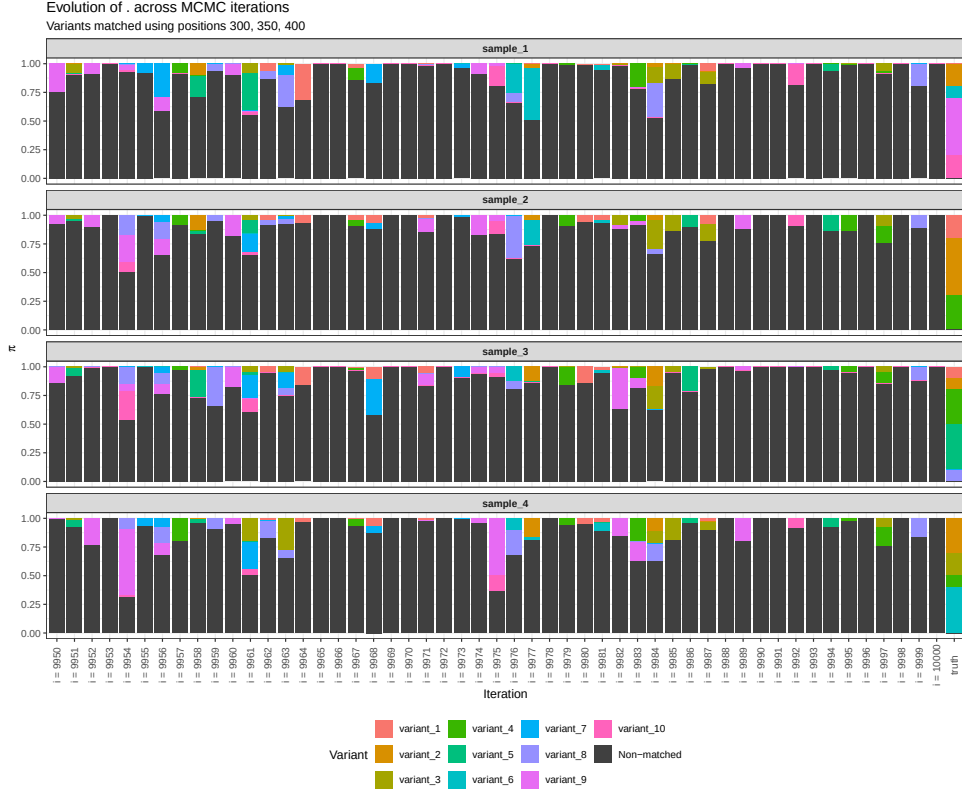


Figure 5: Evolution of the estimated mixture proportions (y-axis) across MCMC iterations (x-axis) in the unknown-variant setting. Each panel corresponds to one biological sample, and each colour represents the estimated proportion associated with one of the true genetic variants after the matching procedure. The final column corresponds to the ground truth proportions used to generate the dataset. The black component represents inferred variants that could not be matched to any of the true variants and therefore correspond to non-matched inferred sequences.

5 Subsampling Methods

In the previous sections, we presented Bayesian inference methods for both the known-variant and unknown-variant settings, where the likelihood was evaluated using the complete dataset at each iteration of the Markov chain. While this approach provides accurate inference, it becomes computationally demanding as the size of the sequencing data increases.

In this section, we revisit both settings and introduce subsampling techniques to reduce computational burden. The main idea is to reduce the cost of likelihood evaluations by replacing the full dataset with a reduced or aggregated representation, either through read subsampling, position subsampling, or exact aggregation of identical read classes, thereby significantly decreasing the computational cost of simulating the Markov chain. The objective is to obtain reliable approximations of the full-data posterior distribution while substantially reducing execution times. The following sections investigate the performance of these subsampling methods in both the known-variant and unknown-variant scenarios.

5.1 Subsampling Strategies

Before introducing the different read subsampling strategies, we define two equivalence relations on the read space that will be used throughout this section.

The first relation groups reads sharing the same sample and the same set of covered genomic positions. More precisely, for $r, r' \in \mathcal{R}$, we define

$$r \sim r' \iff (s_r, \mathbf{v}_r) = (s_{r'}, \mathbf{v}_{r'}).$$

The corresponding equivalence classes are denoted by $\tilde{\mathcal{R}} = \mathcal{R} / \sim$.

For each class $\tilde{r} \in \tilde{\mathcal{R}}$, we define its multiplicity $m_{\tilde{r}} = |\{r \in \mathcal{R} : r \sim \tilde{r}\}|$ and $n_{\tilde{r}} = |\mathbf{v}_{\tilde{r}}|$,

which corresponds to the number of genomic positions covered by any representative read of the class.

A finer partition is obtained by additionally requiring reads to share the observed nucleotides at every covered position. We therefore define

$$r \approx r' \iff (s_r, \mathbf{v}_r, X_r) = (s_{r'}, \mathbf{v}_{r'}, X_{r'}),$$

where X_r denotes the observed sequence associated with read r .

The corresponding equivalence classes are denoted by $\tilde{\tilde{\mathcal{R}}} = \mathcal{R} / \approx$.

For each class $\tilde{\tilde{r}} \in \tilde{\tilde{\mathcal{R}}}$, we use the same notation $m_{\tilde{\tilde{r}}}$ and $n_{\tilde{\tilde{r}}}$ to denote its multiplicity and the number of covered genomic positions, respectively.

These finer equivalence classes play a key role in the likelihood computation. Since all reads within the same class have identical contributions to the likelihood, this quantity only needs to be evaluated once per equivalence class and then weighted by its multiplicity, thereby avoiding redundant computations.

5.1.1 Read Subsampling

Using the equivalence classes introduced above, the subsampling strategies considered in this work operate by selecting a subset of classes rather than individual reads. Once the selected classes have been determined, all reads belonging to these classes are retained. The multiplicities $m_{\tilde{r}}$ or $m_{\tilde{\tilde{r}}}$ are preserved and used in the likelihood computation.

Uniform Subsampling. The simplest strategy consists of retaining a fixed proportion $\rho \in (0, 1]$ of the equivalence classes uniformly at random. More precisely, if

$$N = |\tilde{\mathcal{R}}|,$$

we sample uniformly a subset

$$\tilde{\mathcal{R}}_0 \subset \tilde{\mathcal{R}}, \quad |\tilde{\mathcal{R}}_0| = \lfloor \rho N \rfloor.$$

All reads belonging to the selected classes are retained in the reduced dataset.

Coverage-Based Subsampling. This strategy favors equivalence classes that cover a large number of genomic positions while penalizing reads that mainly cover positions already covered by other selected reads. The objective is to retain long and informative reads, ensuring that all genomic positions remain sufficiently covered while reducing redundant information. For each class $\tilde{r} \in \tilde{\mathcal{R}}$, we define the score

$$\text{score}(\tilde{r}) = \sum_{v \in \mathbf{v}_{\tilde{r}}} \frac{1}{1 + c(v)},$$

where $c(v)$ denotes the number of reads covering position v in the complete dataset.

The sampling probabilities are then defined by

$$p(\tilde{r}) = \frac{\text{score}(\tilde{r})}{\sum_{\tilde{r}' \in \tilde{\mathcal{R}}} \text{score}(\tilde{r}')},$$

and a proportion ρ of the equivalence classes is sampled according to this probability distribution.

Additional subsampling strategies have also been investigated but are not included in this report. These methods, namely **Frequency-Based Subsampling** F.1 and **Hierarchical Subsampling** F.2, are described in Appendix F.1 and Appendix F.2, respectively.

Final remark After selecting a subset of reads $\mathcal{R}_0$ using one of the subsampling strategies described above, the corresponding observed data are denoted by $X^{(\mathcal{R}_0)}$. The likelihood restricted to this subset can be written as:

$$\mathbb{P}\left(X^{(\mathcal{R}_0)} \mid \tau, \varepsilon, \pi\right) = \prod_{r \in \mathcal{R}_0} \sum_{g=1}^G \pi_{g,s_r} \prod_{v \in \mathbf{v}_r} \prod_{a \in \mathcal{A}} \left(\sum_{b \in \mathcal{A}} \tau_{g,v,b} \varepsilon_{b,a} \right)^{X_{r,v,a}}.$$

5.1.2 Position Subsampling

We consider a preprocessing step aimed at reducing the dimensionality of the dataset by selecting informative genomic positions. This procedure is performed both at the level of individual samples and on the aggregated dataset.

Let $n_{v,s,a}$ denote the number of occurrences of nucleotide $a \in \mathcal{A}$ at position $v \in \mathcal{V}$ within sample s , and let

$$n_{v,a} = \sum_s n_{v,s,a}$$

denote the corresponding aggregated count across all samples.

We define the normalised frequencies

$$\hat{p}_{v,s,a} = \frac{n_{v,s,a}}{\sum_{a' \in \mathcal{A}} n_{v,s,a'}}, \quad \hat{p}_{v,a} = \frac{n_{v,a}}{\sum_{a' \in \mathcal{A}} n_{v,a'}}.$$

For each position v , we quantify its variability based on these normalised frequencies and introduce a threshold $\alpha > 0$ to distinguish between true biological signal and sequencing noise. A position v is classified as informative if either of the following conditions holds:

(i) at the aggregated level, there exist at least two distinct nucleotides $a_1 \neq a_2 \in \mathcal{A}$ such that

$$\hat{p}_{v,a_1} > \alpha \quad \text{and} \quad \hat{p}_{v,a_2} > \alpha,$$

or (ii) there exists a sample s such that two distinct nucleotides $a_1 \neq a_2 \in \mathcal{A}$ satisfy

$$\hat{p}_{v,s,a_1} > \alpha \quad \text{and} \quad \hat{p}_{v,s,a_2} > \alpha.$$

We denote by $\mathcal{U}_0 \subset \mathcal{V}$ the resulting set of selected positions, defined according to this criterion. This set includes all informative positions together with a random subset of the remaining ones in order to preserve information about sequencing error patterns.

The dataset is then restricted to observations indexed by $\mathcal{U}_0$, i.e.,

$$X^{(\mathcal{U}_0)} = \{X_{r,v,a} : v \in \mathcal{U}_0\}.$$

Under this restriction, the likelihood defined in Section 2.4 becomes

$$\mathbb{P}\left(X^{(\mathcal{U}_0)} \mid \tau, \varepsilon, \pi\right) = \prod_{r \in \mathcal{R}} \sum_{g=1}^G \pi_{g,s_r} \prod_{v \in \mathbf{v}_r \cap \mathcal{U}_0} \prod_{a \in \mathcal{A}} \left(\sum_{b \in \mathcal{A}} \tau_{g,v,b} \varepsilon_{b,a} \right)^{X_{r,v,a}}.$$

Remark.

This procedure yields variant-specific sequences restricted to the selected set of positions. The remaining positions are not expected to contribute to variant discrimination, as they have been previously identified as largely invariant, with a clearly dominant nucleotide across samples. Consequently, to fully reconstruct the underlying gene sequences, these positions can be resolved independently by assigning the majority nucleotide at each site. This post-processing step is straightforward due to the prior filtering, which significantly reduces the complexity of the data structure.

5.1.3 Mixed Strategies

The subsampling strategies presented above are not mutually exclusive and can be combined to construct more selective sampling procedures. Rather than relying on a single criterion, several strategies may be applied simultaneously, each filtering the dataset from a different perspective.

More precisely, let

$$\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_K$$

denote a collection of subsampling strategies. A mixed strategy consists of successively applying these procedures, producing a reduced dataset

$$\mathcal{R}_0 = (\mathcal{S}_K \circ \dots \circ \mathcal{S}_1)(\mathcal{R}),$$

where the output of one strategy becomes the input of the next.

The motivation behind this approach is that different strategies preserve different aspects of the data. For example, one strategy may ensure a good representation of low-coverage genomic regions, while another preserves the diversity of nucleotide configurations within each equivalence class. By combining several criteria, the resulting subsample can retain complementary sources of information that would not necessarily be preserved by any individual strategy.

The combinations considered in this work are evaluated experimentally in the benchmark section, where their impact on likelihood approximation, posterior estimation and computational efficiency is assessed.

5.2 Likelihood Computation Optimization

Using the equivalence classes introduced previously, the likelihood defined in Section 3.4.4,

$$\mathbb{P}(X \mid \tau, \varepsilon, \pi) = \prod_{r \in \mathcal{R}} \sum_{g=1}^G \pi_{g,s_r} \prod_{v \in \mathbf{v}_r} \prod_{a \in \mathcal{A}} \left(\sum_{b \in \mathcal{A}} \tau_{g,v,b} \varepsilon_{b,a} \right)^{X_{r,v,a}},$$

can be reformulated to avoid repeated evaluations over identical reads.

Indeed, the likelihood contribution of a read depends only on the sample, the covered genomic positions and the observed nucleotides, that is, on the finer equivalence class $\tilde{r} \in \tilde{\mathcal{R}}$.

Therefore, all reads belonging to the same class share the same likelihood contribution, which can be written as

$$\mathcal{L}(\tilde{r}; \tau, \varepsilon, \pi) = \sum_{g=1}^G \pi_{g,s_{\tilde{r}}} \prod_{v \in \mathbf{v}_{\tilde{r}}} \prod_{a \in \mathcal{A}} \left(\sum_{b \in \mathcal{A}} \tau_{g,v,b} \varepsilon_{b,a} \right)^{X_{\tilde{r},v,a}}.$$

Instead of evaluating the likelihood independently for every read, it is therefore sufficient to evaluate it once for each fine equivalence class and weight the result by its multiplicity. The likelihood becomes

$$\mathbb{P}(X \mid \tau, \varepsilon, \pi) = \prod_{\tilde{r} \in \tilde{\mathcal{R}}} [\mathcal{L}(\tilde{r}; \tau, \varepsilon, \pi)]^{m_{\tilde{r}}}.$$

This reformulation is exactly equivalent to the original likelihood and introduces no approximation. It simply exploits the redundancy of the dataset by replacing repeated likelihood evaluations over identical reads with a single evaluation per fine equivalence class, resulting in a substantial reduction in computational cost.

5.3 Benchmark

In this section, we compare the different subsampling methods introduced previously in order to evaluate their computational efficiency and their impact on the likelihood approximation. We consider again a dataset composed of 10 samples and 10 variants, where the variants differ at positions 300, 350, and/or 400.

Recall that $U \subseteq \mathcal{V}$ denotes the subset of genomic positions selected by the subsampling strategy.

The datasets used for the benchmark are summarised in the following table:

Name	Method	U/%	Computational optimization
Data_reads	All Data	-	No
Data_reads_U1	Position subsampling	U1 = [300,400]	No
Data_reads_U2	Position subsampling	U2 = [250,450]	No
Data_reads_U3	Position subsampling	U3 = [200,500]	No
Data_reads_0.25	Coverage subsampling	25%	No
Data_reads_0.5	Coverage subsampling	50%	No
Data_reads_0.75	Coverage subsampling	75%	No
Data_class	All Data	-	Yes
Data_class_U1	Position subsampling	U1 = [300,400]	Yes
Data_class_U2	Position subsampling	U2 = [250,450]	Yes
Data_class_U3	Position subsampling	U3 = [200,500]	Yes
Data_class_0.25	Coverage subsampling	25%	Yes
Data_class_0.5	Coverage subsampling	50%	Yes
Data_class_0.75	Coverage subsampling	75%	Yes

Table 1: Datasets used for the benchmark of the different subsampling strategies.

5.3.1 Accuracy and Computational Performance

To evaluate the proposed likelihood approximation methods, I analyse both their statistical accuracy and their computational efficiency. The corresponding results are presented in Figures 6 and 7.

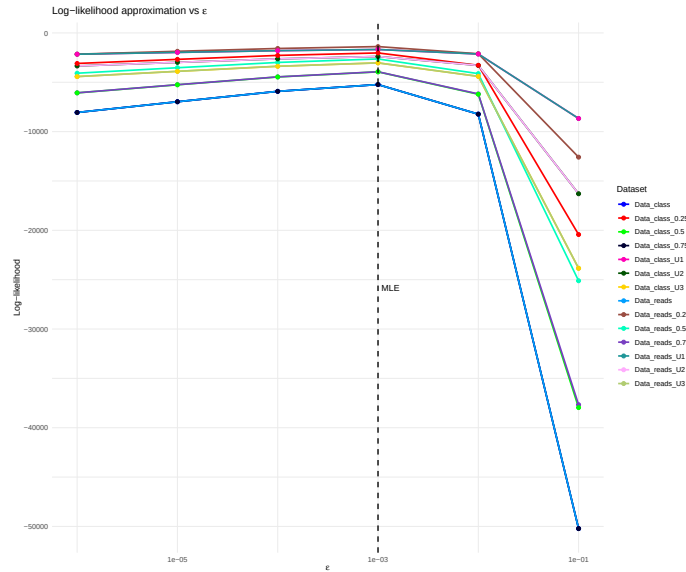


Figure 6: Log-likelihood approximation as a function of the sequencing error rate ϵ for the different subsampling strategies. The x-axis represents the sequencing error rate ϵ (displayed on a logarithmic scale), while the y-axis shows the corresponding approximated log-likelihood. Each coloured curve corresponds to one likelihood approximation method. The dashed vertical line indicates the maximum likelihood estimate (MLE).

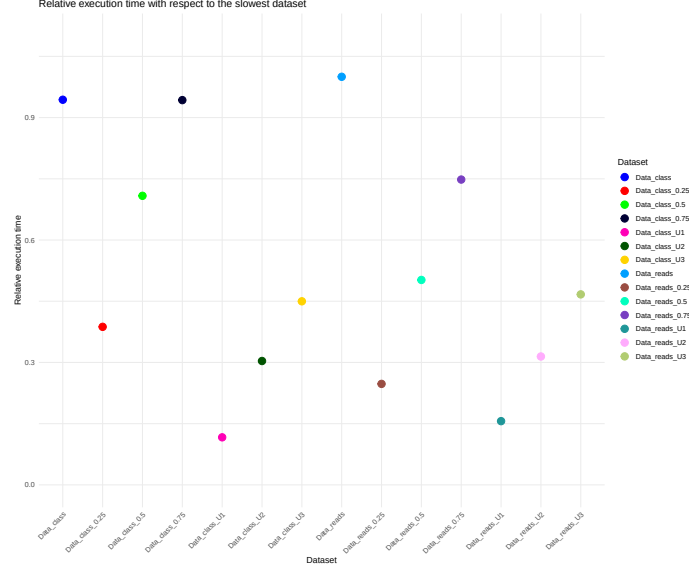


Figure 7: Relative computational performance of the different likelihood approximation methods on real datasets. The x-axis lists the different likelihood approximation methods, while the y-axis represents their execution time relative to the reference method. Each point corresponds to one dataset, allowing the computational performance of the different approaches to be compared across datasets.

The main observations are summarised below: In Figure 6, I compare the approximated log-likelihood as a function of the sequencing error rate ϵ for the different subsampling strategies, keeping the other parameters fixed at the value used for simulating the data. My objective is to verify that the corresponding maximum approximate likelihood estimators remain close to the maximum likelihood estimator while preserving the overall shape of the likelihood function.

- **Statistical accuracy of the likelihood approximation.** In Figure 6, I compare the approximated log-likelihood as a function of the sequencing error rate ϵ for the different subsampling strategies, while keeping the remaining parameters fixed at their true values. I observe that all approximation methods produce very similar likelihood curves and that each has a maximum near the maximum likelihood estimate, indicating that they hopefully preserve the statistical behaviour of the original likelihood.
- **Computational performance.** In Figure 7, I compare the relative execution times of the different likelihood approximation methods on real sequencing datasets. I observe that all the proposed subsampling strategies substantially reduce the computational cost compared with the original likelihood evaluation. In particular, the position-based subsampling methods provide the largest speed-ups while preserving the statistical accuracy observed in Figure 6.

5.3.2 Posterior Distribution Comparison

Following the benchmark results presented above, I observed that the position subsampling strategy provides the best compromise between computational efficiency and likelihood approximation accuracy. Therefore, I decided to further investigate an extreme case by considering only the informative positions included in U4:

$$\{299, 300, 301, 349, 350, 351, 399, 400, 401\}.$$

The objective of this strategy is to retain the most informative positions (300, 350 and 400) while keeping neighbouring positions to preserve sufficient information about the sequencing error rate.

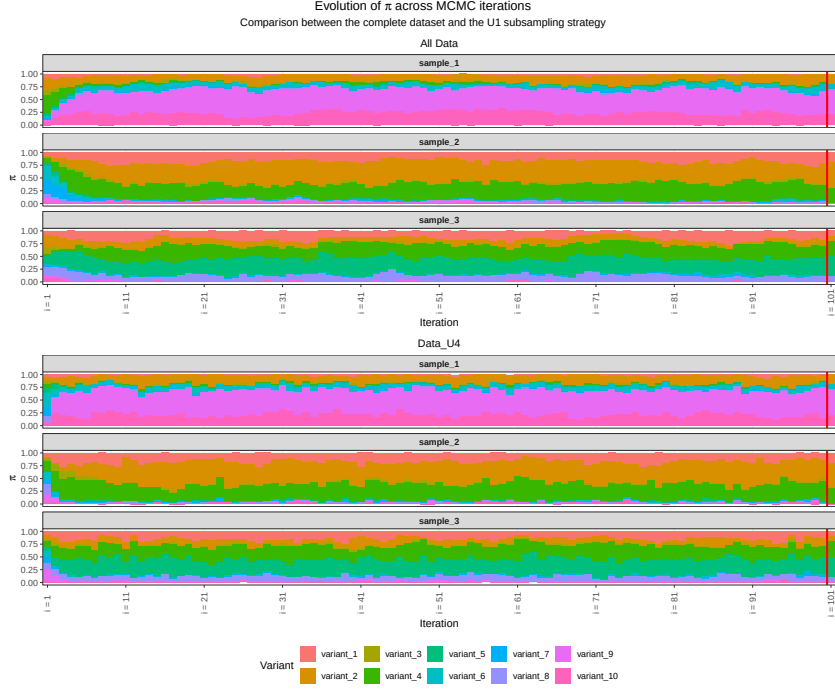


Figure 8: Evolution of the estimated mixture proportions throughout the MCMC iterations. The upper row corresponds to the inference performed using the complete dataset, whereas the lower row corresponds to the inference obtained using only the subset U_4 . Each panel represents one biological sample, and the final column corresponds to the true mixture proportions used to generate the simulated dataset.

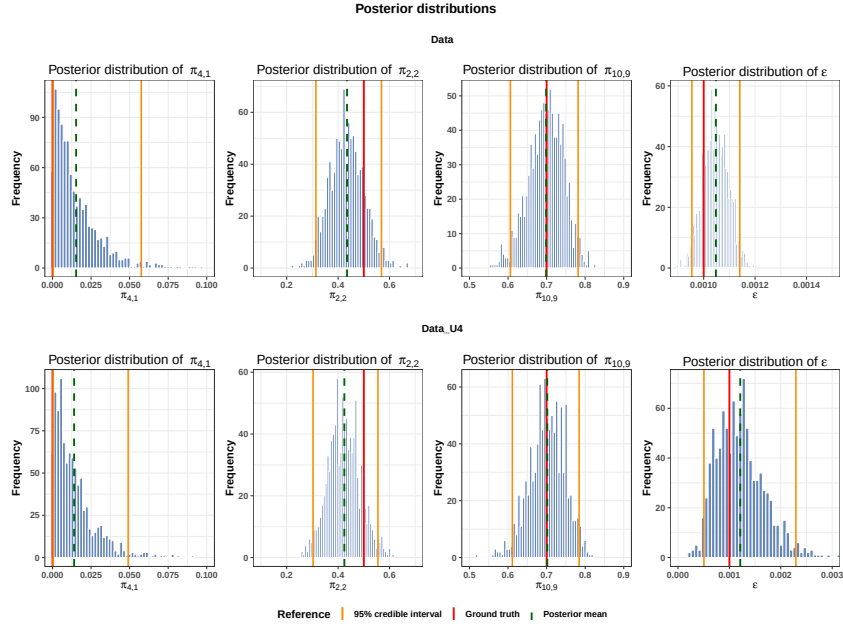


Figure 9: Comparison of the posterior distributions of the sequencing error rate ε together with several representative mixture proportion parameters. The upper row corresponds to the inference performed using the complete dataset, whereas the lower row corresponds to the inference obtained using only the subset U_4 . The vertical lines indicate the posterior means, the 95% credible intervals and the true parameter values.

The main observations are summarised below:

- **Comparison of the estimated mixture proportions.** From Figure 8, I observe that both inference procedures converge towards very similar mixture proportions. Although the estimates obtained using only the subset U_4 exhibit slightly larger fluctuations throughout the MCMC iterations, the overall behaviour remains very similar to that obtained with the complete dataset. This increased variability can be observed, for example, for variant 9 in sample 1, where the chain based on U_4 explores a slightly wider range of values while remaining centred around the same abundance.
- **Comparison of the posterior distributions.** Figure 9 shows that the posterior means and the 95% credible intervals are remarkably similar for the two inference procedures. Moreover, the posterior distributions appear to be centred around the same values, providing strong evidence that the approximation based on U_4 preserves the posterior inference. The main noticeable difference concerns the sequencing error rate ε , for which the posterior distribution obtained using U_4 is slightly more dispersed. This increased uncertainty is expected, since the aggressive reduction in the number of genomic positions inevitably removes information about sequencing errors. Nevertheless, I believe that this additional uncertainty has only a limited impact on the overall inference.

6 Conclusion

The Gibbs sampler developed for the known-variant setting is now fully implemented and provides very encouraging results. In particular, the algorithm accurately estimates the relative proportions of the different variants, represented by the mixture weights π_g , as well as the sequencing error rate ε . Moreover, the proposed likelihood approximation based on subsampling strategies provides a substantial reduction in computational cost while preserving the quality of the posterior estimates across all the methods considered. Although our framework is inspired by the DESMAN model, the probabilistic formulation developed in this work is not a direct implementation of the original method. In particular, the likelihood is formulated differently and several modelling assumptions have been modified with respect to the original model 3.4.7. These results therefore validate both the proposed probabilistic model and the likelihood approximation introduced in this work.

The extension to the unknown-variant setting remains considerably more challenging. As expected, the strong dependency between the latent variant sequences τ and the latent read assignments Z prevents the Gibbs sampler from efficiently exploring the posterior distribution. In particular, the Markov chain tends to become trapped in local modes, limiting posterior exploration and preventing satisfactory convergence.

To overcome these limitations, we have also investigated several alternative sampling strategies. In particular, **Block Sampling** approaches H.1, where several components of τ are updated simultaneously, **Block-wise Sampling** strategies H.2, designed to reduce the computational cost associated with standard block sampling, and **Metropolis–Hastings** updates H.3. In parallel, we also worked on the construction of an unbiased likelihood estimator based on a **Core–Tail Factorisation** approach G.1. It is not yet computationally efficient, but it would enable exact inference via pseudo-marginal Markov Chain Monte Carlo methods. Although there was not enough time within the scope of this internship to thoroughly analyse these methods in the main body of the report, our preliminary experiments suggest that they significantly improve the mixing behaviour of the Markov chain and constitute promising directions for overcoming the limitations identified in this work. Their detailed descriptions are provided in the Appendix.

Overall, these developments constitute a natural continuation of the present work. Further improvements of the sampling strategy, together with a more extensive experimental evaluation of these alternative MCMC methods and the unbiased likelihood approach, represent promising directions for future research.

Beyond the scientific contributions presented in this report, this internship has been an invaluable learning experience from both a technical and a research perspective. Throughout this

project, I significantly improved my programming skills in R, not only by implementing Bayesian inference algorithms, but also by learning how to write more efficient and maintainable code. In particular, I became much more aware of the importance of algorithmic optimisation, for instance by reducing unnecessary loops, improving data structures, and designing computational strategies capable of handling large-scale datasets. I also gained valuable experience in producing clear and informative scientific figures, an essential aspect of communicating complex statistical results.

More importantly, this internship profoundly changed the way I approach scientific problems. My previous mathematical training was largely centred on obtaining exact or deterministic results. In contrast, Bayesian inference requires reasoning under uncertainty, where conclusions are expressed in terms of probabilities and confidence rather than certainty. At first, this way of thinking was unfamiliar, as the objective is no longer to identify a single exact answer, but rather to quantify uncertainty and assess the plausibility of different hypotheses. Learning to interpret posterior distributions, credible intervals and probabilistic statements has therefore been one of the most valuable aspects of this work. I believe that this experience has considerably broadened my perspective on mathematical modelling and statistical inference, and will provide a solid foundation for my future studies and research projects.

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Appendices

A Nomenclature

Notation	Space	Description
$\mathcal{A} = \{A, C, G, T\}$	Set	Set of possible nucleotides representing genetic sequences.
G	$\mathbb{N}$	Number of genetic variants.
S	$\mathbb{N}$	Number of biological samples.
$\mathcal{R}$	Set	Set of sequencing reads, with each $r \in \mathcal{R}$ representing one read.
$\mathcal{V}$	Set	Set of nucleotide positions covered by all reads.
$\mathcal{V}_r$	Set	Set of nucleotide positions covered by read r .
S_r	$\mathbb{N}$	Sample from which read r originates.
$X_{r,v,a}$	$\{0, 1\}$	Indicator equal to 1 if nucleotide a is observed at position v in read r .
$X_{r,v}$	$\{0, 1\}^4$	One-hot encoding of the nucleotide observed at position v in read r using order (A, C, G, T) .
X_r	$\{0, 1\}^{ \mathcal{V}_r \times 4}$	Matrix encoding the nucleotide composition of read r .
X	$\{0, 1\}^{ \mathcal{R} \times \mathcal{V} \times 4}$	Tensor encoding all sequencing reads.
$\tau_{g,v,a}$	$\{0, 1\}$	Indicator equal to 1 if nucleotide a is present at position v in variant g .
$\tau_{g,v}$	$\{0, 1\}^4$	One-hot encoding of the nucleotide at position v in variant g using order (A, C, G, T) .
τ_g	$\{0, 1\}^{ \mathcal{V} \times 4}$	Matrix encoding the complete sequence of variant g .
τ	$\{0, 1\}^{G \times \mathcal{V} \times 4}$	Tensor encoding all genetic variants.
$\pi_{g,s}$	$[0, 1]$	Relative abundance of variant g in sample s .
$\pi_{\cdot,s}$	$[0, 1]^G$	Vector of variant abundances in sample s , with $\sum_{g=1}^G \pi_{g,s} = 1$.
π	$[0, 1]^{G \times S}$	Matrix of variant abundances across all samples.
ε	$[0, 1]$	Sequencing error rate parameter.
Z_r	$\{1, \dots, G\}$	Latent variant assignment of read r .
Z	$\{1, \dots, G\}^{ \mathcal{R} }$	Collection of all latent read assignments.
$x_{r,v}$	$\mathcal{A}$	Observed nucleotide at position v in read r .
$t_{g,v}$	$\mathcal{A}$	True nucleotide of variant g at position v .
M	$\mathbb{N}$	Number of matched observed nucleotides under the current assignments.
E	$\mathbb{N}$	Number of mismatched observed nucleotides under the current assignments.

Table 2: Main notation used throughout this report.

B Log-likelihood and numerical underflow

Read-level log-likelihood under a fixed variant. For a fixed variant g , the log-likelihood of a read r is given by:

$$\log \mathbb{P}(X_r \mid \tau_g, \varepsilon) = \sum_{v \in \mathcal{V}_r} \sum_{a \in \mathcal{A}} X_{r,v,a} \log \left(\sum_{b \in \mathcal{A}} \tau_{g,v,b} \varepsilon_{b,a} \right).$$

Mixture log-likelihood at read level. When marginalising over variants, we obtain:

$$\log \mathbb{P}(X_r \mid \tau, \varepsilon, \pi, s_r) = \log \left(\sum_{g=1}^G \pi_{g,s_r} \mathbb{P}(X_r \mid \tau_g, \varepsilon) \right).$$

This expression cannot be simplified directly due to the logarithm of a sum. To ensure numerical stability, it is rewritten using the log-sum-exp trick:

$$\log \mathbb{P}(X_r \mid s_r) = \log \left(\sum_{g=1}^G \exp(\log \pi_{g,s_r} + \log \mathbb{P}(X_r \mid \tau_g, \varepsilon)) \right).$$

Full dataset log-likelihood. Finally, the log-likelihood of the full dataset is:

$$\log \mathbb{P}(X \mid \tau, \varepsilon, \pi) = \sum_{r \in \mathcal{R}} \log \left(\sum_{g=1}^G \pi_{g,s_r} \mathbb{P}(X_r \mid \tau_g, \varepsilon) \right),$$

which is equivalently computed in a numerically stable way as:

$$\log \mathbb{P}(X \mid \tau, \varepsilon, \pi) = \sum_{r \in \mathcal{R}} \log \left(\sum_{g=1}^G \exp(\log \pi_{g,s_r} + \log \mathbb{P}(X_r \mid \tau_g, \varepsilon)) \right).$$

Complete log-likelihood of the model. In our implementation, the log-likelihood of the model is defined as a function of all parameters:

$$\mathcal{L}(X \mid \pi, \tau, \varepsilon) = \sum_{r \in \mathcal{R}} \log \left(\sum_{g=1}^G \exp(\log \pi_{g,s_r} + \sum_{v \in \mathcal{V}_r} \sum_{a \in \mathcal{A}} X_{r,v,a} \log(\sum_{b \in \mathcal{A}} \tau_{g,v,b} \varepsilon_{b,a})) \right)$$

where:

- π represents the mixture proportions of the variants within each biological sample,
- τ represents the unknown variant sequences,
- ε represents the sequencing error model,
- X denotes the observed sequencing reads.

This is the exact expression used in practice for numerical computation of the model likelihood.

C DESMAN (Quince et al.) model

DESMAN Quince et al. (2017) uses a likelihood that is closely related to the read-level likelihood introduced above, but expressed after aggregation of the sequencing reads. The main difference is that individual reads are no longer kept as the elementary observations. Instead, each read is first decomposed into observations of length one, corresponding to one genomic position covered by the read, and identical terms are then collected according to their genomic position and biological sample.

For each position $v \in \mathcal{V}$, sample $s \in \{1, \dots, S\}$, and nucleotide $a \in \mathcal{A}$, we define the aggregated count

$$N_{v,s,a} = \sum_{r \in \mathcal{R}: s_r=s, v \in \mathcal{V}_r} X_{r,v,a}.$$

Thus, $N_{v,s,a}$ denotes the number of reads originating from sample s that cover position v and report nucleotide a at that position. Equivalently, the collection $(N_{v,s,a})_{a \in \mathcal{A}}$ summarises the nucleotide observations at position v in sample s .

Starting from the read-level likelihood, the DESMAN formulation is obtained by treating each covered position within a read as an independent observation and then replacing the individual indicators $X_{r,v,a}$ by their aggregated counts $N_{v,s,a}$. Since the generating variant of each single-position observation is unknown, the mixture over variants is taken at the level of each position and sample.

The resulting DESMAN likelihood can therefore be written as

$$\mathbb{P}(N \mid \tau, \varepsilon, \pi) = \prod_{v \in \mathcal{V}} \prod_{s=1}^S \prod_{a \in \mathcal{A}} \left(\sum_{g=1}^G \pi_{g,s} \sum_{b \in \mathcal{A}} \tau_{g,v,b} \varepsilon_{b,a} \right)^{N_{v,s,a}}.$$

Equivalently, by grouping the four nucleotide counts at each position and sample into a multinomial contribution, this can be written as

$$\mathbb{P}(N \mid \tau, \varepsilon, \pi) = \prod_{v \in \mathcal{V}} \prod_{s=1}^S \frac{N_{v,s,\cdot}!}{\prod_{a \in \mathcal{A}} N_{v,s,a}!} \prod_{a \in \mathcal{A}} \left(\sum_{g=1}^G \pi_{g,s} \sum_{b \in \mathcal{A}} \tau_{g,v,b} \varepsilon_{b,a} \right)^{N_{v,s,a}},$$

where

$$N_{v,s,\cdot} = \sum_{a \in \mathcal{A}} N_{v,s,a}.$$

This expression has the same emission structure as the read-level likelihood,

$$\sum_{b \in \mathcal{A}} \tau_{g,v,b} \varepsilon_{b,a},$$

but differs in the way the observations are represented. In the read-level model, the likelihood is a product over reads and covered positions, with exponents given by the indicators $X_{r,v,a}$. In DESMAN, these indicators have been summed over all reads from the same sample covering the same position, leading to exponents given by the aggregated nucleotide counts $N_{v,s,a}$. Thus, DESMAN can be viewed as a count-aggregated version of the read-level model, obtained by splitting reads into length-one observations and collecting identical terms.

Compared with the read-level likelihood introduced previously, this considerably reduces the dimensionality of the inference problem and the computational complexity of evaluating the likelihood, but it loses information about colocations¹ or variant alleles, which can be essential for reconstructing variant sequences.

D Mathematical derivations

D.1 Derivation of the Full Conditional Distribution of Z

Demo:

We start from Bayes' rule:

$$p(Z_r = g \mid X, Z_{-r}, \pi, \varepsilon, \tau) = \frac{p(Z_r = g, X, Z_{-r}, \pi, \varepsilon, \tau)}{p(X, Z_{-r}, \pi, \varepsilon, \tau)}.$$

Applying the chain rule of conditional probability to the numerator:

$$p(Z_r = g, X, Z_{-r}, \pi, \varepsilon, \tau) = p(\pi, \varepsilon, \tau) p(Z_{-r} \mid \pi) \mathbf{p}(\mathbf{Z}_r = \mathbf{g} \mid \pi, \mathbf{s}_r) p(X_{-r} \mid Z_{-r}, \tau, \varepsilon) \mathbf{p}(\mathbf{X}_r \mid \mathbf{Z}_r = \mathbf{g}, \tau, \varepsilon).$$

Thus,

$$p(Z_r = g \mid X, Z_{-r}, \pi, \varepsilon, \tau) = \frac{p(\pi, \varepsilon, \tau) p(Z_{-r} \mid \pi) \mathbf{p}(\mathbf{Z}_r = \mathbf{g} \mid \pi, \mathbf{s}_r) p(X_{-r} \mid Z_{-r}, \tau, \varepsilon) \mathbf{p}(\mathbf{X}_r \mid \mathbf{Z}_r = \mathbf{g}, \tau, \varepsilon)}{p(X, Z_{-r}, \pi, \varepsilon, \tau)}.$$

¹Here, colocation information refers to the fact that nucleotides observed on the same read are known to originate from the same underlying DNA fragment, and therefore from the same latent variant.

Hence, all terms not depending on Z_r cancel, giving:

$$p(Z_r = g \mid X, Z_{-r}, \pi, \varepsilon, \tau) \propto \mathbf{p}(\mathbf{Z}_r = \mathbf{g} \mid \pi_{\cdot, s_r}) \mathbf{p}(\mathbf{X}_r \mid \mathbf{Z}_r = \mathbf{g}, \tau_{\mathbf{g}}, \varepsilon).$$

Substituting the prior for Z_r :

$$p(Z_r = g \mid X, Z_{-r}, \pi, \varepsilon, \tau) \propto \pi_{g, s_r} \mathbf{p}(\mathbf{X}_r \mid \tau_{\mathbf{g}}, \varepsilon).$$

By conditional independence of the reads given the parameters and latent assignments, the conditional distribution of Z_r depends on the data only through X_r . Moreover, conditionally on π , Z_r is independent of Z_{-r} . Thus,

$$p(Z_r = g \mid X, \tau, \pi, \varepsilon) = p(Z_r = g \mid X_r, \tau, \pi, \varepsilon).$$

Normalising over all possible values $g' \in \{1, \dots, G\}$:

$$p(Z_r = g \mid X_r, \pi, \varepsilon, \tau) = \frac{\pi_{g, s_r} \mathbf{p}(\mathbf{X}_r \mid \mathbf{Z}_r = \mathbf{g}, \tau_{\mathbf{g}}, \varepsilon)}{\sum_{g'=1}^G \pi_{g', s_r} \mathbf{p}(\mathbf{X}_r \mid \mathbf{Z}_r = \mathbf{g}', \tau_{\mathbf{g}'}, \varepsilon)}.$$

D.2 Derivation of the Full Conditional Distribution of π

Demo:

A Dirichlet distribution satisfies:

$$\text{Dirichlet}(\alpha_1, \dots, \alpha_G) \propto \prod_{g=1}^G \pi_g^{\alpha_g - 1}.$$

Define $Z_{\mathcal{R}_s}$ as the latent assignments corresponding to reads belonging to sample s :

$$Z_{\mathcal{R}_s} = \{Z_r : s_r = s\}.$$

The conditional distribution of $\pi_{\cdot, s}$ is:

$$p(\pi_{\cdot, s} \mid Z, \varepsilon, X, \tau).$$

Using Bayes' rule:

$$p(\pi_{\cdot, s} \mid Z, \varepsilon, X, \tau) = \frac{p(X, Z, \varepsilon, \tau, \pi_{\cdot, s})}{p(X, Z, \varepsilon, \tau)}.$$

Applying the chain rule of conditional probability:

$$p(\pi_{\cdot, s} \mid Z, \varepsilon, X, \tau) = \frac{p(\varepsilon) p(\tau) \mathbf{p}(\pi_{\cdot, s}) \mathbf{p}(\mathbf{Z}_{\mathcal{R}_s} \mid \pi_{\cdot, s}) p(X \mid Z, \varepsilon, \tau)}{p(X, Z, \varepsilon, \tau)}.$$

Keeping only the terms depending on $\pi_{\cdot, s}$:

$$p(\pi_{\cdot, s} \mid Z, \varepsilon, X, \tau) \propto \mathbf{p}(\pi_{\cdot, s}) \mathbf{p}(\mathbf{Z}_{\mathcal{R}_s} \mid \pi_{\cdot, s}).$$

Using the Dirichlet prior:

$$p(\pi_{\cdot, s}) \propto \prod_{g=1}^G \pi_{g, s}^{\alpha_g - 1},$$

and the categorical likelihood:

$$p(Z_{\mathcal{R}_s} \mid \pi_{\cdot, s}) = \prod_{g=1}^G \pi_{g, s}^{n_{g, s}}.$$

Therefore,

$$p(\pi_{\cdot, s} \mid Z, \varepsilon, X, \tau) \propto \prod_{g=1}^G \pi_{g, s}^{\alpha_g - 1} \prod_{g=1}^G \pi_{g, s}^{n_{g, s}} = \prod_{g=1}^G \pi_{g, s}^{\alpha_g + n_{g, s} - 1}.$$

Hence,

$$\pi_{\cdot, s} \mid Z, \varepsilon, X, \tau \sim \text{Dirichlet}(\alpha_1 + n_{1, s}, \dots, \alpha_G + n_{G, s}).$$

D.3 Derivation of the Full Conditional Distribution of ε

Demo:

A Beta distribution with parameters (a, b) has density (up to proportionality):

$$\text{Beta}(a, b) \propto \varepsilon^{a-1} (1 - \varepsilon)^{b-1}.$$

We now derive the full conditional for ε . Starting from Bayes' rule:

$$p(\varepsilon \mid X, Z, \pi, \tau) = \frac{p(\varepsilon, X, Z, \pi, \tau)}{p(X, Z, \pi, \tau)}.$$

Applying the chain rule of conditional probability:

$$p(\varepsilon, X, Z, \pi, \tau) = \mathbf{p}(\varepsilon) p(\pi) p(\tau) p(Z \mid \pi) \mathbf{p}(\mathbf{X} \mid \mathbf{Z}, \tau, \varepsilon).$$

Therefore,

$$p(\varepsilon \mid X, Z, \pi, \tau) \propto \mathbf{p}(\varepsilon) \mathbf{p}(\mathbf{X} \mid \mathbf{Z}, \tau, \varepsilon),$$

since all other terms do not depend on ε .

We now focus on the read likelihood term. From Section 2,

$$p(X \mid Z, \tau, \varepsilon) = \prod_{r \in \mathcal{R}} p(X_r \mid Z_r, \tau, \varepsilon) = \prod_{r \in \mathcal{R}} \prod_{v \in \mathcal{V}_r} \prod_{a \in \mathcal{A}} \left(\sum_{b \in \mathcal{A}} \tau_{Z_r, v, b} \varepsilon_{b, a} \right)^{X_{r, v, a}}.$$

Using M and E defined previously, we can equivalently rewrite this expression as:

$$p(X \mid Z, \tau, \varepsilon) = \left(\frac{\varepsilon}{3} \right)^E (1 - \varepsilon)^M \propto \varepsilon^E (1 - \varepsilon)^M.$$

Substituting $\mathbf{p}(\varepsilon)$ and $\mathbf{p}(\mathbf{X} \mid \mathbf{Z}, \tau, \varepsilon)$:

$$p(\varepsilon \mid X, Z, \pi, \tau) \propto \varepsilon^{a-1} (1 - \varepsilon)^{b-1} (1 - \varepsilon)^M \varepsilon^E.$$

Rearranging terms:

$$p(\varepsilon \mid X, Z, \pi, \tau) \propto \varepsilon^{a+E-1} (1 - \varepsilon)^{b+M-1}.$$

Hence,

$$\varepsilon \mid X, Z, \pi, \tau \sim \text{Beta}(a + E, b + M).$$

E Pseudo-codes

E.1 Known variants

Algorithm 1 Gibbs Sampler for Posterior Inference

Require: Data X , template τ

Require: Hyperparameters a, b, α

Require: Number of iterations T

1: Initialise $Z^{(0)}, \pi^{(0)}, \varepsilon^{(0)}$

2: **for** $t = 1, \dots, T$ **do**

3: **Update latent assignments** Z

4: **for** each read r **do**

5: **for** $g = 1, \dots, G$ **do**

6:

$$w_{r,g} = \pi_{g,s_r}^{(t-1)} p(X_r \mid \tau_g, \varepsilon^{(t-1)})$$

7: **end for**

8: Sample

$$Z_r^{(t)} \sim \text{Categorical}(\text{normalise}(w_{r,1:G}))$$

9: **end for**

10: **Update mixture proportions** π

11: **for** each sample s **do**

12: **for** $g = 1, \dots, G$ **do**

13:

$$n_{g,s} = \sum_r \mathbf{1}(Z_r^{(t)} = g, s_r = s)$$

14: **end for**

15: Sample

$$\pi_{\cdot,s}^{(t)} \sim \text{Dirichlet}(\alpha + n_{1,s}, \dots, \alpha + n_{G,s})$$

16: **end for**

17: **Update error rate** ε

18:

$$M = \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}_r} \mathbf{1}(x_{r,v} = t_{Z_r^{(t)},v})$$

19:

$$E = \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}_r} \mathbf{1}(x_{r,v} \neq t_{Z_r^{(t)},v})$$

20: Sample

$$\varepsilon^{(t)} \sim \text{Beta}(a + E, b + M)$$

21: **end for**

22: **return** $\{Z^{(t)}, \pi^{(t)}, \varepsilon^{(t)}\}_{t=1}^T$

E.2 Unknown variants

Algorithm 2 Gibbs Sampler with Unknown Variant Sequences

```

1: Input: reads  $X$ , number of variants  $G$ , initialisation  $(\pi^{(0)}, Z^{(0)}, \varepsilon^{(0)}, \tau^{(0)})$ 
2: Output: samples  $\{(\pi^{(t)}, Z^{(t)}, \varepsilon^{(t)}, \tau^{(t)})\}_{t=1}^T$ 
3: for  $t = 1, \dots, T$  do
4:   (Same as known-variant case)
5:   Update  $Z^{(t)} \sim p(Z \mid \pi^{(t-1)}, \tau^{(t-1)}, \varepsilon^{(t-1)}, X)$ 
6:   Update  $\pi^{(t)} \sim p(\pi \mid Z^{(t)})$ 
7:   Update  $\varepsilon^{(t)} \sim p(\varepsilon \mid Z^{(t)}, \tau^{(t-1)}, X)$ 
8:   Update of variant sequences  $\tau$ :
9:   for  $g = 1, \dots, G$  do
10:    for  $v \in \mathcal{V}$  do
11:      Compute sufficient statistics:

```

$$n_{g,v} = \sum_{r \in \mathcal{R}} \mathbf{1}\{Z_r^{(t)} = g, v \in \mathcal{V}_r\}$$

```

12:      for  $b \in \mathcal{A}$  do
13:        Compute nucleotide counts:

```

$$m_{g,v,b} = \sum_{r \in \mathcal{R}} \mathbf{1}\{Z_r^{(t)} = g, v \in \mathcal{V}_r, X_{r,v} = b\}$$

```

14:        Compute unnormalised weight:

```

$$w_b = \theta_{g,v,b} (1 - \varepsilon^{(t)})^{m_{g,v,b}} \left(\frac{\varepsilon^{(t)}}{3} \right)^{n_{g,v} - m_{g,v,b}}$$

```

15:      end for
16:      Compute  $(p_{g,v,b})_{b \in \mathcal{A}} = \text{Normalise}(\{w_b\}_{b \in \mathcal{A}})$ 
17:      Sample:

```

$$\tau_{g,v}^{(t)} \sim \text{Categorical}(\{p_{g,v,b}\}_{b \in \mathcal{A}})$$

```

18:    end for
19:  end for
20: end for
21: return  $\{Z^{(t)}, \pi^{(t)}, \varepsilon^{(t)}, \tau^{(t)}\}_{t=1}^T$ 

```

F Supplementary Read Subsampling Strategies

F.1 Frequency-Based Subsampling.

The multiplicity of each equivalence class provides a natural measure of its representativeness within the dataset. Classes are therefore sampled with probabilities proportional to their multiplicities,

$$p(\tilde{r}) = \frac{m_{\tilde{r}}}{\sum_{\tilde{r}' \in \tilde{\mathcal{R}}} m_{\tilde{r}'}}.$$

Consequently, highly represented read patterns are more likely to be retained in the reduced dataset.

F.2 Hierarchical Subsampling.

This strategy exploits the hierarchical structure of the equivalence classes. For each coarse equivalence class

$$\tilde{r} \in \tilde{\mathcal{R}},$$

let

$$m_{\tilde{r}} = |\{r \in \mathcal{R} : r \sim \tilde{r}\}|$$

denote its multiplicity.

Given a retention proportion $\rho \in (0, 1]$, we independently sample without replacement a subset of reads

$$\mathcal{R}_0(\tilde{r}) \subseteq \tilde{r},$$

such that

$$|\mathcal{R}_0(\tilde{r})| = \lfloor \rho m_{\tilde{r}} \rfloor.$$

The resulting subsampled dataset is therefore

$$\mathcal{R}_0 = \bigcup_{\tilde{r} \in \tilde{\mathcal{R}}} \mathcal{R}_0(\tilde{r}).$$

Since the selection is performed independently within each coarse equivalence class, every combination of sample and covered positions remains represented in the subsampled dataset, while the total number of reads is reduced by approximately a factor ρ . The finer equivalence classes $\tilde{r}$ are therefore induced naturally by the sampled reads rather than being selected explicitly.

G Unbiased Likelihood Estimation

G.1 Core–Tail Factorisation

Although the position subsampling strategy presented in the previous section considerably reduces the computational cost of likelihood evaluations, the likelihood must still be evaluated over every retained position. Since this operation is repeated at each iteration of the MCMC algorithm, it remains the dominant computational cost. To further reduce this cost, we exploit the fact that not all genomic positions contribute equally to the likelihood.

Let

$$\mathcal{V} = \mathcal{V}_{\text{core}} \cup \mathcal{V}_{\text{tail}}, \quad \mathcal{V}_{\text{core}} \cap \mathcal{V}_{\text{tail}} = \emptyset,$$

where $\mathcal{V}_{\text{core}}$ denotes the subset of informative positions selected according to the criterion described in Section 5.1.2, and

$$\mathcal{V}_{\text{tail}} = \mathcal{V} \setminus \mathcal{V}_{\text{core}}$$

contains the remaining positions.

For each read r and variant g , we define the variant-specific likelihood contributions over the core and tail position sets as

$$L_{r,g}^{\text{core}}(\tau_g, \varepsilon) = \prod_{v \in \mathcal{V}_r \cap \mathcal{V}_{\text{core}}} \prod_{a \in \mathcal{A}} \left(\sum_{b \in \mathcal{A}} \tau_{g,v,b} \varepsilon_{b,a} \right)^{X_{r,v,a}},$$

and

$$L_{r,g}^{\text{tail}}(\tau_g, \varepsilon) = \prod_{v \in \mathcal{V}_r \cap \mathcal{V}_{\text{tail}}} \prod_{a \in \mathcal{A}} \left(\sum_{b \in \mathcal{A}} \tau_{g,v,b} \varepsilon_{b,a} \right)^{X_{r,v,a}}.$$

The marginal likelihood contribution of read r is then given by the mixture over variants:

$$L_r(\tau, \varepsilon, \pi) = \sum_{g=1}^G \pi_{g,s_r} L_{r,g}^{\text{core}}(\tau_g, \varepsilon) L_{r,g}^{\text{tail}}(\tau_g, \varepsilon).$$

Rather than evaluating $L_{r,g}^{\text{tail}}$ exactly for each variant g , we rewrite it in log-space as

$$L_{r,g}^{\text{tail}} = \exp \left(\sum_{v \in \mathcal{V}_r \cap \mathcal{V}_{\text{tail}}} \log \ell_{r,g,v} \right),$$

where

$$\ell_{r,g,v} = \prod_{a \in \mathcal{A}} \left(\sum_{b \in \mathcal{A}} \tau_{g,v,b} \varepsilon_{b,a} \right)^{X_{r,v,a}}$$

denotes the likelihood contribution of variant g associated with position v on read r .

This representation transforms the product over tail positions for variant g into the exponential of a sum, allowing the use of stochastic estimators. More precisely, let

$$N \sim \text{Poisson}(\lambda),$$

where $\lambda > 0$ controls the average number of sampled terms. Instead of evaluating every summand in the tail, a random subset of contributions is computed and appropriately reweighted according to its sampling probabilities. This yields an unbiased estimator of the variant-specific tail contribution,

$$\widehat{L}_{r,g}^{\text{tail}},$$

satisfying $\mathbb{E}[\widehat{L}_{r,g}^{\text{tail}}] = L_{r,g}^{\text{tail}}$ conditionally on g . The detailed construction of this Poisson estimator is not developed here; the important point is that the estimator is designed so that its expectation equals the exact tail likelihood contribution.

The read likelihood contribution is then estimated by

$$\widehat{L}_r = \sum_{g=1}^G \pi_{g,s_r} L_{r,g}^{\text{core}} \widehat{L}_{r,g}^{\text{tail}},$$

which satisfies $\mathbb{E}[\widehat{L}_r] = L_r$, and the overall likelihood estimator is given by $\widehat{\mathbb{P}}(X \mid \tau, \varepsilon, \pi) = \prod_{r \in \mathcal{R}} \widehat{L}_r$.

The parameter λ determines the trade-off between computational efficiency and statistical accuracy. Increasing λ improves the precision of the approximation at the expense of additional computations, whereas smaller values produce faster likelihood evaluations with a larger Monte Carlo variance.

H Alternative MCMC Sampling Strategies

The Gibbs sampler developed for the unknown-variant setting suffers from poor mixing due to the strong dependence between the latent variant sequences τ and the latent read assignments Z . As a consequence, modifying a single nucleotide at a time often has little effect on the posterior distribution and makes it difficult for the Markov chain to escape from local modes. To overcome this limitation, I investigated several alternative MCMC sampling strategies that update several components of the latent variant sequences simultaneously, either across variants at a fixed position or through larger coordinated moves. By proposing larger coordinated changes to the sequences, these methods increase the probability of reaching regions of higher posterior probability and improve the exploration of the posterior distribution. The following sections briefly present the main ideas behind these approaches.

H.1 Block Sampling

In the block sampling approach, instead of updating each nucleotide $\tau_{g,v}$ independently, we jointly sample the complete vector of variant nucleotides at a given position:

$$\tau_{1:G,v} = (\tau_{1,v}, \dots, \tau_{G,v}).$$

Unlike the single-site Gibbs update presented in the main text, this block update is based on the marginal likelihood, where the latent assignments Z are integrated out. The corresponding conditional distribution is obtained by integrating the contribution of all variants sharing the same position v :

$$P(\tau_{1:G,v} = \mathbf{b} \mid X, \varepsilon, \pi, \tau_{-(v)}) \propto \left(\prod_{g=1}^G \theta_{g,v,b_g} \right) \prod_{r:v \in \mathcal{V}_r} \left(\sum_{g=1}^G \pi_{g,s_r} L_{r,g}^{-(v)} \prod_{a \in \mathcal{A}} (\varepsilon_{b_g,a})^{X_{r,v,a}} \right),$$

where $\mathbf{b} = (b_1, \dots, b_G) \in \mathcal{A}^G$, $\tau_{-(v)}$ denotes all variant nucleotides except those at position v , and

$$L_{r,g}^{-(v)} = \prod_{v' \in \mathcal{V}_r \setminus \{v\}} P(X_{r,v'} \mid \tau_{g,v'}, \varepsilon).$$

This update allows the nucleotides of all variants at position v to be sampled jointly, therefore accounting for the dependencies induced by the read likelihood and avoiding independent updates of highly correlated variables.

H.2 Block-wise Sampling

A compromise between updating all variants jointly and updating each variant independently is obtained through a block-wise sampling strategy. Similarly to full block sampling, this procedure operates on the marginal likelihood with Z integrated out. At each iteration, the set of variants $\{1, \dots, G\}$ is randomly partitioned into M disjoint blocks:

$$\mathcal{B}^{(t)} = \{B_1^{(t)}, \dots, B_M^{(t)}\},$$

such that

$$\bigcup_{m=1}^M B_m^{(t)} = \{1, \dots, G\}, \quad B_i^{(t)} \cap B_j^{(t)} = \emptyset \quad \text{for } i \neq j.$$

Each block contains a fixed number of variants,

$$|B_m^{(t)}| = K, \quad K < G,$$

where K is chosen as a compromise between computational cost and the ability to capture dependencies between variants. The partition $\mathcal{B}^{(t)}$ is randomly generated at each iteration of the Markov chain, preventing the algorithm from being restricted to a fixed grouping of variants.

For each block $B_m^{(t)}$, the corresponding variables $\tau_{B_m^{(t)},v} = (\tau_{g,v})_{g \in B_m^{(t)}}$ are proposed while keeping the non-block variants fixed at their current nucleotide assignments $\tau_{g,v}$. The variables are sampled from their conditional posterior distribution:

$$P(\tau_{B_m^{(t)},v} = \mathbf{b} \mid X, \varepsilon, \pi, \tau_{-B_m^{(t)},v}) \propto \left(\prod_{g \in B_m^{(t)}} \theta_{g,v,b_g} \right) \prod_{r:v \in \mathcal{V}_r} \left(\sum_{g=1}^G \pi_{g,s_r} L_{r,g}^{-(v)} \prod_{a \in \mathcal{A}} (\varepsilon_{\tilde{b}_g,a})^{X_{r,v,a}} \right),$$

where $\mathbf{b} = (b_g)_{g \in B_m^{(t)}} \in \mathcal{A}^K$, and $\tilde{b}_g = b_g$ for $g \in B_m^{(t)}$ while $\tilde{b}_g = t_{g,v}$ (the current nucleotide) for $g \notin B_m^{(t)}$.

This procedure is repeated for every block until all variants have been updated. Compared with full block sampling, this approach reduces the computational complexity while still allowing the sampler to account for part of the strong dependencies between the latent variables $\tau_{g,v}$.

We also explored Metropolis–Hastings algorithms as an alternative to Gibbs updates. Preliminary experiments suggest that these methods may provide a more efficient exploration of the posterior distribution, with satisfactory acceptance rates and improved movement through the state space, leading to better mixing properties of the Markov chain.

Remark. Although the full-block sampling approach allows joint exploration of the entire space of possible variant configurations at each position, its computational cost increases exponentially with the number of variants in terms of the number of nucleotide configurations evaluated. Indeed, for each position v , the sampler must evaluate all possible nucleotide assignments:

$$|\mathcal{A}|^G = 4^G,$$

leading to a complexity of $\mathcal{O}(4^G)$ configuration evaluations per position (or $\mathcal{O}(4^G \times V)$ across the gene), which quickly becomes intractable as G increases.

The block-wise sampling strategy provides a compromise between computational efficiency and the ability to capture dependencies between variants. Although it does not explore the entire configuration space simultaneously, it provides a good approximation by jointly updating smaller subsets of variants. For a fixed block size K , the number of configurations evaluated per block is reduced to 4^K . Since approximately G/K blocks are required to update all variants, the resulting complexity in terms of configuration evaluations becomes:

$$\mathcal{O}\left(4^K \times \frac{G}{K} \times V\right),$$

which remains computationally feasible for moderate values of K while preserving part of the dependency structure lost in single-site Gibbs sampling.

When $G \not\equiv 0 \pmod{K}$, the last block will contain the remaining $G \bmod K$ variants, resulting in a smaller final block than the others.

H.3 Metropolis-Hastings Sampling

In the Metropolis-Hastings approach, instead of evaluating the conditional distribution over all possible configurations $\mathbf{b} \in \mathcal{A}^G$, we propose a new candidate configuration $\mathbf{b}'$ from a proposal distribution $q(\mathbf{b}' | \mathbf{b})$ under the marginal likelihood formulation and decide whether to accept it.

For a given position v , let the current state be:

$$\mathbf{b} = (b_1, \dots, b_G) \in \mathcal{A}^G,$$

and let

$$\mathbf{b}' = (b'_1, \dots, b'_G) \in \mathcal{A}^G$$

denote a proposed new configuration. The unnormalised posterior probability is computed for both configurations using:

$$\tilde{P}(\mathbf{b}) = \left(\prod_{g=1}^G \theta_{g,v,b_g} \right) \prod_{r: v \in \mathcal{V}_r} \left(\sum_{g=1}^G \pi_{g,s_r} L_{r,g}^{-(v)} \prod_{a \in \mathcal{A}} (\varepsilon_{b_g,a})^{X_{r,v,a}} \right),$$

and similarly for the proposed state $\mathbf{b}'$:

$$\tilde{P}(\mathbf{b}') = \left(\prod_{g=1}^G \theta_{g,v,b'_g} \right) \prod_{r: v \in \mathcal{V}_r} \left(\sum_{g=1}^G \pi_{g,s_r} L_{r,g}^{-(v)} \prod_{a \in \mathcal{A}} (\varepsilon_{b'_g,a})^{X_{r,v,a}} \right).$$

In general, the acceptance probability is given by:

$$\alpha = \min \left(1, \frac{\tilde{P}(\mathbf{b}') q(\mathbf{b} \mid \mathbf{b}')}{\tilde{P}(\mathbf{b}) q(\mathbf{b}' \mid \mathbf{b})} \right).$$

For a symmetric proposal distribution satisfying $q(\mathbf{b}' \mid \mathbf{b}) = q(\mathbf{b} \mid \mathbf{b}')$, this simplifies to:

$$\alpha = \min \left(1, \frac{\tilde{P}(\mathbf{b}')}{\tilde{P}(\mathbf{b})} \right).$$

The proposed configuration $\mathbf{b}'$ is then accepted with probability α ; otherwise, the sampler remains in the current state $\mathbf{b}$.

Towards the end of the internship, we also began investigating hybrid strategies that combine these three approaches with the single-site Gibbs sampling strategy introduced in Section 4.3. Instead of relying on a single sampling mechanism, different transition kernels were applied at different iterations of the Markov chain, allowing the sampler to leverage the strengths of each method. Although these preliminary experiments were promising, there was not enough time to include a complete analysis in this report. A systematic study of these hybrid samplers represents a natural direction for future work.

Finally, an important perspective would be to derive and sample directly from the joint conditional distribution of (Z, τ) . In the current Gibbs sampler, these two variables are updated sequentially, despite their strong correlation. Therefore, the conditional independence assumptions implicitly used by these successive updates may not fully capture the model's dependency structure. Developing an efficient strategy to jointly sample (Z, τ) would provide a more faithful representation of the posterior distribution and could significantly improve both the convergence speed and the mixing properties of the Markov chain.